

Use Case Scenario

Table 1

Authentication

Use Case 1: Authentication

Use Case Goal: To allow authorized users to securely access GAOIRS according to their assigned role.

Scenario: Successful login and access to the assigned dashboard.

Actor: Citizen (Reporter) / Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition: The system is running and accessible. The user has an active registered account.

Main Flow:

Step 1: The user navigates to the GAOIRS login page.

Step 2: The user enters their registered email and password.

Step 3: The user clicks the Login button.

Step 4: The system validates the credentials against the user records in the database.

Step 5: The system determines the user's assigned role (Resident, Response Unit, or Administrator).

Step 6: The system creates an authenticated session and redirects the user to the appropriate dashboard based on their role.

User Action: Enters registered email and password, then clicks Login.

System Response: Validates the credentials, determines the user's role, creates the session, and redirects the user to the appropriate role-based dashboard.

Table 2

Submit Incident Report

Use Case 2: Submit Incident Report

Use Case Goal: To allow a resident to submit a new incident report containing GPS coordinates, incident type, description, and photo evidence.

Scenario: Resident successfully submits an incident report to the system.

Actor: Citizen (Reporter)

Pre-condition: The citizen is logged in. The system is running and accessible.

Main Flow:

Step 1: The resident navigates to the Submit Incident Report form.

Step 2: The resident selects the incident type from the available categories.

Step 3: The system automatically captures the GPS coordinates from the resident's device.

Step 4: The resident enters a description of the incident.

Step 5: The resident uploads photo evidence.

Step 6: The resident clicks the Submit button.

Step 7: The system validates all required fields.

Step 8: The system saves the incident report, assigns a unique incident code, and sets the status to Reported.

Step 9: The system sends a submission confirmation to the resident.

User Action: Fills in the incident details (type, description, GPS location, photo evidence) and clicks Submit.

System Response: Validates all fields, saves the incident report, generates a unique incident code, sets status to Reported, and sends a submission confirmation to the resident.

Table 3

Select Incident Type

Use Case 3: Select Incident Type

Use Case Goal: To allow the resident to categorize the incident by selecting from predefined incident types (e.g., fire, road accident, crime, public disturbance).

Scenario: Resident selects the appropriate incident type during report submission.

Actor: Citizen (Reporter)

Pre-condition: The citizen is on the incident report submission form.

Main Flow:

Step 1: The system displays a list of predefined incident types with their corresponding color-coded labels.

Step 2: The resident selects the incident type that best describes the reported event.

Step 3: The system updates the form to reflect the selected type and applies the corresponding map marker color.

User Action: Selects the incident type from the available list.

System Response: Updates the form with the selected incident type and applies the corresponding color-coded map marker setting.

Table 4

Tag GPS Location

Use Case 4: Tag GPS Location

Use Case Goal: To capture the precise GPS coordinates of the incident location for accurate geospatial mapping and nearest responder detection.

Scenario: GPS coordinates are successfully tagged to the incident report.

Actor: Citizen (Reporter)

Pre-condition: The resident's device has location services enabled. The resident is on the incident report form.

Main Flow:

Step 1: The system requests access to the resident's device GPS.

Step 2: The resident grants location permission.

Step 3: The system retrieves the current GPS coordinates (latitude and longitude).

Step 4: The system displays the pinned location on the interactive map preview within the form.

Step 5: The resident confirms or manually adjusts the pin location if needed.

Step 6: The system stores the confirmed GPS coordinates with the incident report.

User Action: Grants location permission and confirms or adjusts the pinned location on the map preview.

System Response: Retrieves GPS coordinates, displays the location on the map preview, and stores the confirmed coordinates with the incident report.

Table 5

Upload Photo for Evidence

Use Case 5: Upload Photo for Evidence

Use Case Goal: To allow the resident to attach photo evidence to support and validate the submitted incident report.

Scenario: Resident successfully uploads a photo as supporting evidence.

Actor: Citizen (Reporter)

Pre-condition: The resident is filling out the incident report submission form.

Main Flow:

Step 1: The resident clicks the Upload Photo button on the incident report form.

Step 2: The system opens the file picker or camera access on the resident's device.

Step 3: The resident selects or captures the photo.

Step 4: The system validates the file type and file size.

Step 5: The system attaches the photo to the incident report as supporting evidence.

Step 6: The system displays a thumbnail preview of the uploaded photo on the form.

User Action: Clicks the Upload Photo button and selects or captures the photo from the device.

System Response: Validates the file type and size, attaches the photo to the incident report as evidence, and displays a thumbnail preview.

Table 6

Track Report Status

Use Case 6: Track Report Status

Use Case Goal: To allow the resident to monitor the current status of their submitted incident report throughout the response workflow.

Scenario: Resident successfully views the current status and timeline of a submitted report.

Actor: Citizen (Reporter)

Pre-condition: The resident has previously submitted at least one incident report and is logged in.

Main Flow:

Step 1: The resident navigates to the My Reports section.

Step 2: The system retrieves and displays the list of the resident's submitted incident reports.

Step 3: The resident selects a specific report to view.

Step 4: The system displays the current status of the incident (Reported, Verified, Responding, Resolved, or Closed).

Step 5: The system shows the incident status timeline with timestamps for each stage.

User Action: Navigates to the My Reports section and selects a specific report to view its status.

System Response: Retrieves and displays the selected report's current status and a timestamped status timeline.

Table 7

Receive Submission Confirmation

Use Case 7: Receive Submission Confirmation

Use Case Goal: To notify the resident that their incident report has been successfully received and logged in the system.

Scenario: Resident receives a confirmation notification after successfully submitting an incident report.

Actor: Resident (Reporter)

Pre-condition: The resident has successfully submitted an incident report.

Main Flow:

Step 1: The system processes and saves the submitted incident report.

Step 2: The system generates a unique incident reference code.

Step 3: The system sends a confirmation notification to the resident via the system dashboard and/or email.

Step 4: The resident views the confirmation message along with the assigned incident reference code.

User Action: Views the submission confirmation message on the dashboard.

System Response: Generates a unique incident reference code and sends a submission confirmation notification to the resident via the dashboard and/or email.

Table 8

Acknowledge Incident

Use Case 8: Acknowledge Incident

Use Case Goal: To allow the response unit to formally acknowledge receipt of a dispatched incident assignment.

Scenario: Response unit successfully acknowledges an assigned incident.

Actor: Response Unit (BFP/PNP/DRRMO)

Pre-condition: An incident has been reported, verified, and dispatched to the response unit. The response unit is logged in.

Main Flow:

Step 1: The system sends an automated dispatch alert to the nearest available response unit.

Step 2: The response unit receives the incident notification on their dashboard.

Step 3: The response unit reviews the incident details, GPS location, and photo evidence.

Step 4: The response unit clicks the Acknowledge button.

Step 5: The system records the acknowledgment timestamp and updates the incident assignment status.

Step 6: The system notifies the administrator and the reporter that the incident has been acknowledged.

User Action: Reviews the incident details and clicks Acknowledge.

System Response: Records the acknowledgment timestamp, updates the assignment status, and notifies both the administrator and the reporter.

Table 9

Update Incident Status

Use Case 9: Update Incident Status

Use Case Goal: To allow the response unit to update the current status of an active incident during response operations.

Scenario: Response unit updates the incident status while responding to the scene.

Actor: Response Unit (BFP/PNP/DRRMO)

Pre-condition: An incident has been acknowledged, and the response unit is actively responding. The response unit is logged in.

Main Flow:

Step 1: The response unit navigates to the assigned incident record.

Step 2: The response unit selects the new status from the available options (Responding, Resolved, or Closed).

Step 3: The response unit optionally adds remarks or operational notes.

Step 4: The response unit clicks Update Status.

Step 5: The system validates and saves the new status with the timestamp.

Step 6: The system logs the status change in the Incident Status Log.

Step 7: The system notifies the reporter and administrator of the updated status.

User Action: Selects a new incident status, optionally adds remarks, and clicks Update Status.

System Response: Saves the updated status with timestamp, logs the change in the Incident Status Log, and notifies the reporter and administrator.

Table 10

Resolve / Close Incident

Use Case 10: Resolve / Close Incident

Use Case Goal: To allow the response unit to formally mark an incident as resolved after completing on-site response operations.

Scenario: Response unit marks an incident as resolved after completing response operations.

Actor: Response Unit (BFP/PNP/DRRMO)

Pre-condition: The incident is in Responding status. The response unit has completed on-site operations and is logged in.

Main Flow:

Step 1: The response unit opens the active incident record.

Step 2: The response unit uploads the response report and/or evidence (optional).

Step 3: The response unit selects Resolved from the status options.

Step 4: The response unit enters resolution remarks.

Step 5: The response unit clicks Update Status.

Step 6: The system updates the incident status to Resolved and records the resolution timestamp.

Step 7: The system logs the resolution in the Incident Status Log.

Step 8: The system notifies the reporter and administrator that the incident has been resolved.

User Action: Uploads response evidence, selects Resolved, enters resolution remarks, and clicks Update Status.

System Response: Updates the incident status to Resolved, records the resolution timestamp, logs the change, and notifies the reporter and administrator.

Table 11

View Interactive Map

Use Case 11: View Interactive Map

Use Case Goal: To allow the response unit and administrator to view all reported incidents plotted on the interactive geospatial map with color-coded markers.

Scenario: Response unit views the interactive map to locate active incidents and plan response routes.

Actor: Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition: The user is logged in. At least one incident has been reported in the system.

Main Flow:

Step 1: The user navigates to the Map View section.

Step 2: The system loads the interactive geospatial map and plots all reported incidents using color-coded markers by incident type.

Step 3: The system applies geofencing to display incidents within the correct barangay/jurisdictional zone.

Step 4: The user applies filters (incident type, status, date range, or barangay zone) to refine the map view.

Step 5: The user clicks on a specific incident marker to view the incident details popup.

Step 6: The system displays the incident information including type, description, GPS location, status, and reporter details.

User Action: Navigates to the Map View, applies filters, and clicks on incident markers to view details.

System Response: Loads the interactive map with geofenced, color-coded incident markers and displays detailed incident information on marker click.

Table 12

Upload Response Report / Evidence

Use Case 12: Upload Response Report / Evidence

Use Case Goal: To allow the response unit to attach the official response report and supporting evidence to the incident record after completing response operations.

Scenario: Response unit uploads a response report and evidence upon incident resolution.

Actor: Response Unit (BFP/PNP/DRRMO)

Pre-condition: The incident has been responded to. The response unit is logged in.

Main Flow:

Step 1: The response unit opens the assigned incident record.

Step 2: The response unit clicks the Upload Response Report/Evidence button.

Step 3: The response unit selects the file(s) to upload (photos, documents).

Step 4: The system validates the file type and size.

Step 5: The system attaches the file(s) to the incident record under Supporting Details.

Step 6: The system confirms the upload and updates the incident record.

User Action: Selects and uploads the response report and/or evidence files.

System Response: Validates and attaches the uploaded files to the incident record as supporting details and confirms the upload.

Table 13

Manage Incidents

Use Case 13: Manage Incidents

Use Case Goal: To allow authorized users to view, filter, and manage all incident records in the system.

Scenario: Administrator reviews and manages all active and resolved incident records.

Actor: Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition: The user is logged in with incident management privileges. Incidents have been reported in the system.

Main Flow:

Step 1: The user navigates to the Incident Management section.

Step 2: The system displays a list of all incident records with their type, status, date, barangay zone, and severity.

Step 3: The user applies filters (type, status, date range, barangay zone) to narrow the incident list.

Step 4: The user selects a specific incident to view its full details.

Step 5: The user updates the incident status, assigns a response unit, or adds administrative remarks as needed.

Step 6: The system saves all changes and logs the action in the Incident Status Log.

User Action: Filters the incident list, selects an incident, and updates status, assignment, or remarks as needed.

System Response: Displays filtered incident records, saves any updates, and logs all changes in the Incident Status Log.

Table 14

Dispatch Response Team

Use Case 14: Dispatch Response Team

Use Case Goal: To assign the nearest available response unit to a reported incident and dispatch them via automated alerts.

Scenario: System detects the nearest available responder, and the administrator dispatches the response team.

Actor: Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition: An incident has been reported and verified. Response units have updated availability status in the system.

Main Flow:

Step 1: The system computes the nearest available response unit based on the incident's GPS coordinates and unit availability.

Step 2: The system displays the nearest responder recommendation on the administrator's dashboard.

Step 3: The administrator reviews the recommendation and confirms or manually selects a different response unit.

Step 4: The system sends an automated dispatch alert to the selected unit via the dashboard, SMS, and/or email.

Step 5: The response unit receives the dispatch order containing incident details, GPS location, and map data.

Step 6: The system records the dispatch timestamp and updates the incident status to Responding.

User Action: Reviews the nearest responder recommendation and confirms or manually selects the response unit for dispatch.

System Response: Sends automated dispatch alerts to the selected response unit, records the dispatch timestamp, and updates the incident status to Responding.

Table 15

Generate Incident Report

Use Case 15: Generate Incident Report

Use Case Goal: To allow authorized users to generate and export incident summary, area-based, or response time analysis reports in PDF, Excel, or CSV format.

Scenario: Administrator generates a monthly incident summary report filtered by incident type and barangay zone.

Actor: Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition:

The user is logged in with report generation privileges. Incident data is available in the system.

Main Flow:

Step 1: The user navigates to the Report Generation section.

Step 2: The user selects the report type (incident summary, area-based report, or response time analysis).

Step 3: The user applies filters (date range, incident type, barangay zone, or status).

Step 4: The user selects the export format (PDF, Excel, or CSV).

Step 5: The user clicks Generate Report.

Step 6: The system compiles the filtered incident data and generates the report.

Step 7: The system makes the report available for download and saves a record in Generated Reports.

User Action: Selects the report type, applies filters, chooses the export format, and clicks Generate Report.

System Response: Compiles the filtered incident data, generates the report in the selected format, provides it for download, and saves it in Generated Reports.

Table 16

View Analytics Dashboard

Use Case 16: View Analytics Dashboard

Use Case Goal: To allow authorized users to view real-time incident analytics, heatmaps, hotspot visualizations, and trend analysis to support evidence-based planning and resource allocation.

Scenario: Administrator views the analytics dashboard to analyze incident trends and identify hotspot areas.

Actor: Response Unit (BFP/PNP/DRRMO) / Administrator

Pre-condition: The user is logged in with dashboard access. Incident records are available in the system.

Main Flow:

Step 1: The user navigates to the Analytics Dashboard.

Step 2: The system loads and displays incident counts by type, status, barangay zone, and time period.

Step 3: The system renders heatmap and hotspot visualizations on the geospatial map.

Step 4: The user applies filters (date range, incident type, barangay zone) to refine the analysis view.

Step 5: The system updates all charts, graphs, and map visualizations in real time based on applied filters.

Step 6: The user reviews temporal and spatial incident patterns to support planning and resource allocation decisions.

User Action: Navigates to the Analytics Dashboard and applies filters to explore incident trends, hotspots, and distribution data.

System Response: Loads and updates all analytics visualizations including charts, heatmaps, and hotspot maps based on the applied filters.

Table 17

Manage User Accounts

Use Case 17: Manage User Accounts

Use Case Goal:

To allow the administrator to create, update, activate, or deactivate user accounts across all roles in the system.

Scenario: Administrator creates a new user account for response unit personnel.

Actor: Administrator

Pre-condition: The administrator is logged in with full system access.

Main Flow:

Step 1: The administrator navigates to the User Management section.

Step 2: The administrator clicks Add New User.

Step 3: The administrator fills in the user's name, email, contact number, and assigns a role (admin, responder, or citizen).

Step 4: The administrator assigns the user to a barangay and/or response unit as applicable.

Step 5: The system validates the input and creates the new user account.

Step 6: The system sends the login credentials to the new user's registered email.

Step 7: The administrator can also edit, deactivate, or delete existing accounts from the user list.

User Action: Fills in the new user's details, assigns the role and barangay/unit, and clicks Save.

System Response:

Validates and creates the new user account, then sends login credentials to the user's registered email address.

Table 18

Manage Incident Types / RBAC

Use Case 18: Manage Incident Types / RBAC

Use Case Goal: To allow the administrator to configure incident type categories and manage role-based access control (RBAC) settings system-wide.

Scenario: Administrator adds a new incident type and reviews role-based access permissions.

Actor: Administrator

Pre-condition: The administrator is logged in with system configuration privileges.

Main Flow:

Step 1: The administrator navigates to the System Configuration section.

Step 2: The administrator opens the Incident Types management panel.

Step 3: The administrator adds, edits, or removes incident types, setting the name, color code, and icon label for each.

Step 4: The administrator navigates to the Role-Based Access Control (RBAC) settings panel.

Step 5: The administrator reviews the permissions currently assigned to each role (admin, responder, citizen).

Step 6: The administrator updates role permissions as needed.

Step 7: The system saves the updated configuration and applies the changes system-wide.

User Action: Configures incident type details and updates role permissions, then saves the changes.

System Response: Saves the updated incident type configurations and RBAC permission settings and applies them across the system.
